Roll Number: ____________

Course:	Introduction to Data Science	Course Code:	DS 2001
Program:	BS(DS)	Semester:	Fall 2022
Duration:	1 Hour	Total Marks:	40
Paper Date:	26-09-2022	Page(s):	8
Section:	BS (DS) A, B, C, D	Section:	
Exam:	Mid I	Roll No:	

Instructions: Answer in the space provided. You can ask for rough sheets, but they will not be graded or marked. In case of confusion or ambiguity make a reasonable assumption. Questions during exam are not allowed.

Question # 1: **[15 Marks]**

Suppose that you are working with a healthcare company HealTech. HealTech hired you to work on a project that understands the impact of age and weight on the person's overall stamina to work. The initial dataset given to you for analysis is as follows:

Age (years)	Weight (Kg)	Stamina to work
36	71	11
33	65	13
29	61	14
27	55	16
25	51	20

Your job is to train an appropriate model, and get the values for the required parameters. Once the values are known, predict the "stamina to work" if the person's age is 24 years and weight is 45 kg.

Using Least Square Method.

Question 1

$$\sum X_1 = N \cdot a_1 + b_{12\cdot3} \sum X_2 + b_{13\cdot2} \sum X_3$$

$$\sum X_1 X_2 = a_{1\cdot23} \sum X_2 + b_{12\cdot3} \sum (X_2)^2 + b_{13\cdot2} \sum X_2 X_3$$

$$\sum X_1 X_3 = a_{1\cdot23} \sum X_3 + b_{12\cdot3} \sum X_2 X_3 + b_{13\cdot2} \sum (X_3)^2$$

	Age	weight					
X_1	X_2	X_3	X_1X_2	X_1X_3	X_2X_3	X_2^2	X_3^2
11	36	71	396	781	2556	1296	5041
13	33	65	429	845	2145	1089	4225
14	29	61	406	854	1769	841	3721
16	27	55	432	880	1485	729	3025
20	25	51	500	1020	1275	625	2601
74	150	303	2163	4380	9230	4580	18613

Now, Equation becomes

$$74 = 5a_{1\cdot23} + b_{12\cdot3}\,150 + b_{13\cdot2}\,303$$

i.e

$$5a_{1\cdot23} + 150\,b_{12\cdot3} + 303\,b_{13\cdot2} = 74 \quad \text{— (4)}$$

$$2163 = a_{1\cdot23}\,150 + b_{12\cdot3}\,4580 + b_{13\cdot2}\,9230$$

i.e

$$150a_{1\cdot23} + 4580\,b_{12\cdot3} + 9230\,b_{13\cdot2} = 2163 \quad \text{— (5)}$$

$$4380 = a_{1\cdot23}\,303 + b_{12\cdot3}\,9230 + b_{13\cdot2}\,18613$$

i.e.

$$303\,a_{1\cdot23} + 9230\,b_{12\cdot3} + 18613\,b_{13\cdot2} = 4380 \quad \text{— (6)}$$

Now Eq. (4) × 30, becomes.

$$150 a_{1.23} + 4500 b_{12.3} + 9090 b_{13.2} = 2220 \quad \text{— (7)}$$

Now Subtract (7) − (5)

$$150 a_{1.23} + 4500 b_{12.3} + 9090 b_{13.2} = 2220$$
$$\pm 150 a_{1.23} \pm 4580 b_{12.3} \pm 9230 b_{13.2} = \pm 2163$$

$$0 - 80 b_{12.3} - 140 b_{13.2} = +57$$

$$80 b_{12.3} + 140 b_{13.2} = -57 \quad \text{— (8)}$$

Now Eq. (4) × 303 and Eq. (6) × 5. becomes.

$$1515 a_{1.23} + 45450 b_{12.3} + 91809 b_{13.2} = 224$$
$$\pm 1515 a_{1.23} \pm 46150 b_{12.3} \pm 93065 b_{13.2} = \pm 21900$$

$$0 - 700 b_{12.3} - 1256 b_{13.2} = 522$$

$$700 b_{12.3} + 1256 b_{13.2} = -522 \quad \text{— (}$$

Multiplying 80 * (9) and 700 * (8)

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$$56000\, b_{12\cdot 3} + 98000\, b_{13\cdot 2} = -39900$$

$$\pm\; 56000\, b_{12\cdot 3} \;\overset{+}{-}\; 100480\, b_{13\cdot 2} = \overset{-}{+}\; 41780$$

$$0 \quad - 2480\, b_{13\cdot 2} = +1880$$

$$b_{13\cdot 2} = \frac{1880}{-2480}$$

$$\boxed{b_{13\cdot 2} = -0.758}$$

Put $b_{13\cdot 2}$ in eq. (8)

$$80\, b_{12\cdot 3} + 140(b_{13\cdot 2}) = -57$$

$$80\, b_{12\cdot 3} + 140(-0.758) = -57$$

$$80\, b_{12\cdot 3} - 106.12 = -57$$

$$80\, b_{12\cdot 3} = -57 + 106.12$$

$$80\, b_{12\cdot 3} = 49.12$$

$$b_{12\cdot 3} = \frac{49.12}{80}$$

$$\boxed{b_{12\cdot 3} = 0.614}$$

Put $b_{13\cdot2}$ and $b_{12\cdot3}$ in eq. (4)

$$5a_{12\cdot3} + 150\, b_{12\cdot3} + 303\, b_{13\cdot2} = 74$$

$$5a_{12\cdot3} + 150(0.614) + 303(-0.758) = 74$$

$$5a_{12\cdot3} + 92.1 - 229.674 = 74$$

$$5a_{12\cdot3} = 74 - 92.1 + 229.674$$

$$5a_{12\cdot3} = 211.574$$

$$a_{12\cdot3} = \frac{211.574}{5}$$

$$\boxed{a_{12\cdot3} = 42.3}$$

Regression line:

$$\hat{Y} = a + bX_1 + bX_2$$

$$= 42.3 + b_{12\cdot3} X_2 + b_{13\cdot2} X_3$$

$$= 42.3 + 0.614 X_2 - 0.758 X_3$$

$$\hat{Y} = 42.3 + 0.614(45) - 0.758(24)$$

$$\hat{Y} = 42.3 + 27.63 - 18.192$$

$$\boxed{\hat{Y} = 51.738}$$

Question # 2: **[5 Marks]**

While working for HealTech, a task assigned to you is to determine the quality of the machine learning model trained on the blood pressure dataset. In order to determine the quality, the dataset contains only predicted and actual blood pressure values.

Compute the required values, and comment on the machine learning model's performance.

Actual Blood Pressure	Predicted Blood Pressure
78	79
80	85
100	105
105	101
96	93
91	85

Y	Y-pred	(y-Ymean)	(y-Ymean) ^2->SST	(Y-pred-Ymean)	(Y-pred-Ymean)^2->SSR
78	79	-13.66666667	186.7777778	-12.66666667	160.4444444
80	85	-11.66666667	136.1111111	-6.666666667	44.44444444
100	105	8.333333333	69.44444444	13.33333333	177.7777778
105	101	13.33333333	177.7777778	9.333333333	87.11111111
96	93	4.333333333	18.77777778	1.333333333	1.777777778
91	85	-0.666666667	0.444444444	-6.666666667	44.44444444
91.66667			**589.3333333**		**516**

R^2= SSR/SST	0.875565611

Regression relationship is strong as it is 87.5%

Question # 3: **[5 + 5 Marks]**

Given an input function J(θ) = θ^3, using gradient decent to find θ that minimizes cost. The initial value is θ = 3, and the learning rate (α) = 0.2. Compute the first 5 values. Further, write a python code (outline only) for the gradient descent algorithm for this case specifically (also mention the stop criteria/ convergence point.

Iteration	θ	$\alpha \frac{d}{d\theta} J(\theta)$
1	3	5.4
2	-2.4	3.456
3	-5.856	20.5756
4	-26.4316	419.1790
5	-445.609	119140.43

$$\alpha \frac{d}{d\theta} J(\theta)$$

- ⇨ **(0.2)*3ϴ^2**
- ⇨

Code:

```
theeta_old=0
theeta_new=3
eps=0.2
convergence=0.001
```

```
def theetaFunction(theeta):
    return 3* theeta**2

while abs(theeta_new-theeta_old)>convergence:
    theeta_old=theeta_new
    theeta_new= theeta_old-eps*theetaFunction(theeta_old)
```

Question # 4: [10 Marks]

Implement simple linear regression using sklearn in Python. You are required to import the dataset into your file and perform data preprocessing steps. Use 75% of the total dataset in the training of your model. The screenshot of first twenty rows of your dataset "titanic.csv" is attached for your reference. The dataset might have some missing values but Fare for each ticket is given. Your goal is to predict whether the passenger survived the sinking of the Titanic or not. Moreover, you are required to plot the best fit line on your dataset. You can remove unnecessary features if any.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, M	male	22	1	0	A/5 21171	7.25		S
2	1	1	Cumings, M	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen,	female	26	0	0	STON/O2.	7.925		S
4	1	1	Futrelle, M	female	35	1	0	113803	53.1	C123	S
5	0	3	Allen, Mr. '	male	35	0	0	373450	8.05		S
6	0	3	Moran, Mr	male		0	0	330877	8.4583		Q
7	0	1	McCarthy,	male	54	0	0	17463	51.8625	E46	S
8	0	3	Palsson, M	male	2	3	1	349909	21.075		S
9	1	3	Johnson, M	female	27	0	2	347742	11.1333		S
10	1	2	Nasser, Mr	female	14	1	0	237736	30.0708		C
11	1	3	Sandstrom	female	4	1	1	PP 9549	16.7	G6	S
12	1	1	Bonnell, M	female	58	0	0	113783	26.55	C103	S
13	0	3	Saunderco	male	20	0	0	A/5. 2151	8.05		S
14	0	3	Andersson	male	39	1	5	347082	31.275		S
15	0	3	Vestrom, M	female	14	0	0	350406	7.8542		S
16	1	2	Hewlett, M	female	55	0	0	248706	16		S
17	0	3	Rice, Mast	male	2	4	1	382652	29.125		Q
18	1	2	Williams, M	male		0	0	244373	13		S
19	0	3	Vander Pla	female	31	1	0	345763	18		S
20	1	3	Masselmai	female		0	0	2649	7.225		C

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn import linear_model
```

```
#Read the csv file:

data = pd.read_csv("titanic.csv")
data = data[["Fare","Survived"]]
data.head()

#Taking 75% of the data for training and 20% for testing:
num = int(len(data)*0.75)

#Training data:
train = data[:num]

#Testing data:
test = data[num:]

print ("Data: ",len(data))
print ("Train: ",len(train))
print ("Test: ",len(test))

#Training the model:

regr = linear_model.LinearRegression()

train_x = np.array(train[["Fare"]])
train_y = np.array(train[["Survived"]])

regr.fit(train_x,train_y)

coefficients = regr.coef_
intercept = regr.intercept_

#Slope:
print ("Slope: ",coefficients[0])

#Intercept:
print("Intercept: ",intercept)

#Plotting the regression line with train data:

plt.scatter(train["Fare"], train["Survived"])
plt.plot(train_x, coefficients[0]*train_x + intercept,color="red")
```

```
plt.xlabel("Fare")
plt.ylabel("Survived")
plt.show()

predicted_test = regr.predict(test[["Fare"]])
predicted_test[0:5]

#Plot the regression line for testing data:

plt.scatter(test["Fare"],test["Survived"])
plt.plot(test["ENGINESIZE"],predicted_test,color="red")
plt.xlabel("Fare")
plt.ylabel("Survived")
plt.show()
```

Good Luck!